MODULE 1: CHEMICAL THERMODYNAMICS AND KINETICS

Complete Unabridged Course Notes, Derivations, Diagrams & Topic-Wise CAT/FAT Solved Problems

Fully Aligned with Official VIT Faculty Lecture Decks (58 Slides), Department Question Bank, and Past FAT Papers (CO: 1)

Module 1 Detailed Syllabus Breakdown (100% Complete Coverage):

Part A: Thermodynamics: Terminologies (Systems, Surroundings, Boundaries, State/Path Functions), Zeroth Law & Applications, First Law, Enthalpy & Heat Capacity (C_p, C_v), Mayer's Relation Derivation, Work done in Isothermal, Adiabatic, Isobaric, and Isochoric processes, Fire Extinguisher Effect, Second Law (Kelvin-Planck & Clausius statements), Entropy (ΔS in reversible/irreversible processes), Carnot Cycle & Thermal Efficiency Derivation, Refrigerator COP, Free Energy (Helmholtz A and Gibbs G), Gibbs-Helmholtz Equation Derivation, Spontaneity Criteria, Third Law of Thermodynamics & Residual Entropy.
Part B: Chemical Kinetics: Reaction Rates, Differential Rate Expressions, Order vs. Molecularity, Integrated Rate Equations (0th, 1st, 2nd Order), Half-Life Calculations, Rate Plots, Pseudo-First-Order Reactions, Collision Theory & Steric Factor, Activation Energy (E_a), Reaction Coordinate Diagrams, Arrhenius Equation & Derivations, Catalysis (Homogeneous vs Heterogeneous, Lowering of E_a, Industrial Processes).

PART A: CHEMICAL THERMODYNAMICS

1. Basic Terminologies and Fundamental Concepts

Thermodynamics is the branch of physical science dealing with the quantitative relationships between heat, work, and other forms of energy in chemical and physical transformations.

1.1 Thermodynamic Systems, Surroundings, and Boundaries

- **System:** That specific part of the universe selected for thermodynamic study or observation (e.g., reacting chemicals in a flask).
- **Surroundings:** The entire rest of the universe outside the boundaries of the system that can interact with it.
- **Boundary:** The real or imaginary surface separating the system from its surroundings. Boundaries may be:
  - **Rigid or Flexible:** Rigid boundaries cannot move ($\Delta V = 0$, no boundary work); flexible boundaries allow volume changes ($\Delta V \neq 0$).
  - **Diathermal or Adiabatic:** Diathermal boundaries permit heat transfer ($dq \neq 0$); adiabatic boundaries are perfectly thermally insulated ($dq = 0$).
  - **Permeable or Impermeable:** Permeable allows mass flow; impermeable prevents mass transfer.

System Type	Mass Exchange?	Energy Exchange?	Physical Example
Open System	Yes ($\Delta m \neq 0$)	Yes ($\Delta E \neq 0$)	Open beaker with boiling water; biological cells
Closed System	No ($\Delta m = 0$)	Yes ($\Delta E \neq 0$)	Gas in a closed cylinder with movable piston; sealed beverage can
Isolated System	No ($\Delta m = 0$)	No ($\Delta E = 0$)	Liquid in a perfectly insulated vacuum Dewar flask; the Universe as a whole

1.2 State Functions vs. Path Functions

- **State Functions (Variables of State):** Properties that depend solely on the current thermodynamic state of the system, independent of the path or sequence of steps taken to reach that state.
  - Examples: Pressure (P), Volume (V), Temperature (T), Internal Energy (U), Enthalpy (H), Entropy (S), Gibbs Free Energy (G).
  - Mathematical property: Cyclic integral is zero: $\oint dX = 0$; change is $\Delta X = X_{\text{final}} - X_{\text{initial}}$.
- **Path Functions:** Quantities whose magnitudes depend strictly on the particular path or mechanism followed during the transition.
  - Examples: Heat absorbed or evolved (q), Work performed (w).
  - Mathematical property: Inexact differentials: $\oint dq \neq 0$, $\oint dw \neq 0$.

1.3 Intensive vs. Extensive Properties

- **Intensive Properties:** Values are independent of the total mass, size, or quantity of matter in the system.
  - Examples: Temperature (T), Pressure (P), Density ($\rho = \frac{m}{V}$), Refractive index, Boiling pintegral.cont, Viscosity, Surface tension, Molar heat capacity (C_m).
- **Extensive Properties:** Values are directly additive and proportional to the total mass or quantity of matter in the system.
  - Examples: Mass (m), Volume (V), Total Internal Energy (U), Enthalpy (H), Entropy (S), Heat capacity (C).
  - **Key Rule:** The ratio of any two extensive properties is an intensive property (e.g., $\frac{m}{V}$ = Density).

1.4 Thermodynamic Processes

- **Isothermal Process:** Conducted at strictly constant temperature ($dT = 0, \Delta T = 0$). For an ideal gas, $\Delta U = 0$ and $\Delta H = 0$.
- **Adiabatic Process:** Perfectly thermally insulated system where no heat enters or leaves ($dq = 0, q = 0$).
- **Isobaric Process:** Carried out at constant pressure ($dP = 0, \Delta P = 0$). Work done is $w = -P\Delta V$.
- **Isochoric Process:** Carried out at constant volume ($dV = 0, \Delta V = 0$). Expansion work is zero ($w = 0$).
- **Cyclic Process:** System undergoes a series of operations and returns exactly to its initial state ($\Delta U = 0, \Delta H = 0$).

1.5 Reversible vs. Irreversible Processes

Parameter	Reversible Process	Irreversible Process
Driving Force vs Opposing Force	Driving force exceeds opposing force by an infinitesimal amount (dP).	Driving force is significantly greater than opposing force ($\Delta P \gg 0$).
Equilibrium	Thermodynamic equilibrium is maintained at every intermediate stage.	Equilibrium exists only at the initial and final states.
Time Required	Proceeds infinitely slowly; takes infinite time to reach completion.	Proceeds at a measurable, finite speed.
Work Obtained	Yields the maximum possible expansion work (w_{max}).	Yields substantially less work due to dissipative losses (friction, turbulence).
Direction Reversal	Can be reversed at any pintegral.cont by an infinitesimal change in external conditions.	Cannot be reversed without leaving permanent changes in the surroundings.
Natural Occurrence	Ideal, hypothetical construct; cannot be fully realized in practice.	All real, spontaneous natural processes are strictly irreversible.

[CAT-1 Exam Question — 5 Marks]
Question: Differentiate between reversible and irreversible processes with at least five distinct comparative pintegral.conts and suitable examples.
Model Answer Breakdown:
1. **Definition & Force:** In reversible, $P_{ext} = P_{int} \pm dP$ (driving and opposing forces differ infinitesimally). In irreversible, $P_{ext} \ll P_{int}$ or $P_{ext} \gg P_{int}$. (1 Mark)
2. **Equilibrium State:** Reversible processes pass through continuous quasi-static equilibrium states; irreversible processes do not. (1 Mark)
3. **Speed:** Reversible takes infinite time; irreversible occurs in finite time. (1 Mark)
4. **Work Output:** Reversible expansion produces maximum work ($|w_{rev}| > |w_{irrev}|$). (1 Mark)
5. **Examples:** Reversible: Frictionless piston expansion with infinitesimal sand grain removal. Irreversible: Free expansion of gas into a vacuum, heat flow from hot to cold body, neutralization of strong acid and strong base. (1 Mark)

2. The Zeroth Law of Thermodynamics

Formal Statement of Zeroth Law:
"If two physical bodies (A and B) are each in thermal equilibrium with a third body (C), then they are in mutual thermal equilibrium with each other."

- **Concept of Thermal Equilibrium:** Two systems are in thermal equilibrium when no net thermal energy flows between them when placed in diathermal contact.
- **Physical Significance:** The Zeroth Law introduces the concept and objective definition of **Temperature** as a thermodynamic scalar state variable. If $T_A = T_C$ and $T_B = T_C$, then $T_A = T_B$.
- **Industrial & Practical Applications:**
  1. **Temperature Measurement (Thermometry):** A thermometer acts as "Body C". When placed in contact with system A, it reaches equilibrium, establishing a reading. When placed in system B and shows the same reading, $T_A = T_B$.
  2. **Calibration Standards:** Used to calibrate clinical thermometers, thermocouples, resistance temperature detectors (RTDs), and optical pyrometers against standard fixed pintegral.conts (e.g., the triple pintegral.cont of water at 273.16 K, 0.01° C, and 611.65 Pa).

3. The First Law of Thermodynamics and Enthalpy

Formal Statement of First Law (Conservation of Energy):
"Energy can neither be created nor destroyed; it can only be transformed from one form to another. The total energy of an isolated system remains strictly constant."

3.1 Mathematical Formulation and Sign Conventions (IUPAC)

$$\Delta U = q + w$$

where:
- $\Delta U = U_2 - U_1$ is the change in internal energy (State Function, in Joules).
- q is heat absorbed by the system ($q > 0$: endothermic; $q < 0$: exothermic).
- w is work done on the system ($w > 0$: compression; $w < 0$: expansion).
- For an infinitesimal change: $dU = dq + dw = dq - P_{ext}dV$.

3.2 Enthalpy (H)

Enthalpy represents the total heat content of a thermodynamic system at constant pressure, defined as:

$$H = U + PV$$

Taking the differential: $dH = dU + PdV + VdP$. At constant pressure ($dP = 0$):

$$dH = dU + PdV$$

From the First Law, $dU = dq - PdV$, hence:

$$dH = (dq - PdV) + PdV = dq_p$$

Key Theorem: The heat exchanged at constant pressure is equal to the change in enthalpy ($q_p = \Delta H$), whereas the heat exchanged at constant volume is equal to the change in internal energy ($q_v = \Delta U$).

3.3 Molar Heat Capacities (C_p and C_v)

- **Heat Capacity (C):** The quantity of heat required to raise the temperature of a system by 1 Kelvin: $C = d\frac{q}{d}T$.
- **At Constant Volume (C_v):** Since $dq_v = dU$,

$$C_v = \left(\frac{\partial U}{\partial T}\right)_V$$

- **At Constant Pressure (C_p):** Since $dq_p = dH$,

$$C_p = \left(\frac{\partial H}{\partial T}\right)_P$$

3.4 Derivation of Mayer's Relation: $C_p - C_v = R$

For 1 mole of an ideal gas:

1. By definition, enthalpy is: $H = U + PV$.
2. For an ideal gas, the equation of state is: $PV = RT$.
3. Substituting $PV = RT$:

$$H = U + RT$$

4. Differentiating both sides with respect to temperature T at constant pressure:

$$\left(\frac{\partial H}{\partial T}\right)_P = \left(\frac{\partial U}{\partial T}\right)_P + R$$

5. For an ideal gas, internal energy U depends solely on temperature (Joule's Law: $\left(\partial \frac{U}{\partial} V\right)_T = 0$), so $\left(\partial \frac{U}{\partial} T\right)_P = \left(\partial \frac{U}{\partial} T\right)_V = C_v$.
6. Since $\left(\partial \frac{H}{\partial} T\right)_P = C_p$, substitution gives:

$$C_p = C_v + R \Longrightarrow C_p - C_v = R$$

Physical Reason why $C_p > C_v$:
At constant volume, all absorbed heat is utilized exclusively to increase internal energy (ΔU) and raise temperature. At constant pressure, the gas expands against external atmospheric pressure; thus, absorbed heat must not only raise the internal energy (ΔU) but also perform external $P\Delta V$ work. Hence, extra heat equivalent to R is required per mole per Kelvin.

[CAT-1 / FAT Exam Derivation — 5 Marks]
Question: Prove that for 1 mole of an ideal gas, $C_p - C_v = R$. Explain why the molar heat capacity at constant pressure is always greater than that at constant volume.
Step-by-Step Model Scoring Guide:

- State definitions $H = U + PV$ and $PV = RT$: (1 Mark)
- Substitute to obtain $H = U + RT$: (1 Mark)
- Differentiate with respect to T at constant P to yield $\left(\partial \frac{H}{\partial} T\right)_P = \left(\partial \frac{U}{\partial} T\right)_P + R$: (1 Mark)
- Identify $C_p = \left(\partial \frac{H}{\partial} T\right)_P$ and apply Joule's law $\left(\partial \frac{U}{\partial} T\right)_P = C_v$: (1 Mark)
- Conclude $C_p - C_v = R$ and provide physical justification regarding expansion work: (1 Mark)

4. Application of First Law to Expansion Work

4.1 Reversible Isothermal Expansion of an Ideal Gas

For an isothermal process of an ideal gas, $T = \text{constant} \Longrightarrow \Delta U = 0$. From the First Law: $q = -w$. Work done during an expansion from V_1 to V_2:

$$w_{\text{rev}} = -\int_{V_1}^{V_2} P_{\text{ext}} dV = -\int_{V_1}^{V_2} \left(\frac{nRT}{V}\right) dV = -nRT \ln\left(\frac{V_2}{V_1}\right)$$

Converting to base-10 logarithm and using Boyle's Law ($P_1V_1 = P_2V_2 \Longrightarrow \frac{V_2}{V_1} = \frac{P_1}{P_2}$):

$$w_{\text{rev}} = -2.303nRT \log_{10}\left(\frac{V_2}{V_1}\right) = -2.303nRT \log_{10}\left(\frac{P_1}{P_2}\right)$$

4.2 Irreversible Isothermal Expansion Against Constant External Pressure

If gas expands suddenly against a constant opposing pressure P_{ext}:

$$w_{\text{irrev}} = -P_{\text{ext}}(V_2 - V_1) = -P_{\text{ext}}\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right)$$

Note: Since $P_{\text{ext}} < P_{\text{int}}$ during expansion, $|w_{\text{rev}}| > |w_{\text{irrev}}|$. Reversible work represents the maximum theoretical work.

4.3 Free Expansion into Vacuum ($w_{\text{vacuum}} = 0$)

When a gas expands into an evacuated chamber, the external opposing pressure is strictly zero ($P_{\text{ext}} = 0$).

$$w = -\int 0 \cdot dV = 0$$

If the expansion is also adiabatic ($q = 0$), then from First Law $\Delta U = q + w = 0$. Since $\Delta U = nC_v\Delta T = 0$, $\Delta T = 0$ (no temperature change for an ideal gas).

4.4 Reversible Adiabatic Expansion

In an adiabatic process, no heat enters or leaves the system: $dq = 0$. From First Law: $dU = dw = -PdV$. Since $dU = nC_v dT$, we have:

$$nC_v dT = -\left(\frac{nRT}{V}\right) dV \Longrightarrow \frac{C_v}{T} dT = -\frac{R}{V} dV$$

Integrating between states (T_1, V_1) and (T_2, V_2) and setting $\gamma = \frac{C_p}{C_v}$ ($R = C_p - C_v$):

$$\ln\left(\frac{T_2}{T_1}\right) = -(\gamma - 1) \ln\left(\frac{V_2}{V_1}\right) \Longrightarrow T_1 V_1^{\gamma - 1} = T_2 V_2^{\gamma - 1}$$

Substituting $T = P\frac{V}{n}R$:

$$P_1 V_1^{\gamma} = P_2 V_2^{\gamma} = \text{constant}$$

Work Done in Adiabatic Expansion:

$$w_{\text{adiabatic}} = \Delta U = nC_v(T_2 - T_1) = \frac{nR(T_2 - T_1)}{\gamma - 1} = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$$

Since $V_2 > V_1$ in an expansion, $T_2 < T_1$ (temperature drops; the system cools down because it performs boundary work at the expense of its own internal energy).

4.5 Real-World Case Study: The Fire Extinguisher Effect

- Inside a CO_2 fire extinguisher, carbon dioxide is stored under high pressure ($\approx 55 - 60$ bar) in liquid-vapor equilibrium.
- When the valve is opened, the gas rushes out through the nozzle into atmospheric pressure (1 bar).
- This expansion occurs so rapidly that heat exchange with the surrounding air is negligible ($dq \approx 0$, adiabatic expansion).

- The gas performs extensive mechanical work pushing against atmospheric pressure at the expense of its own internal kinetic energy.
- The sudden temperature plunge ($\Delta T \ll 0$) drops the temperature well below the sublimation pintegral.cont (-78.5° C), immediately converting the gaseous CO_2 into solid dry ice snow flakes, creating an effective cooling and oxygen-displacing blanket.

[CAT-1 Numerical Problem — 5 Marks]
Problem: 2 moles of an ideal gas at 300 K and 10 atm expand isothermally and reversibly to a final pressure of 2 atm. Calculate (i) the work done (w), (ii) heat absorbed (q), and (iii) change in internal energy (ΔU). ($R = 8.314 \text{J K}^{-1}\text{mol}^{-1}$).
Step-by-Step Solution:
1. **Change in Internal Energy (ΔU):** For an isothermal process of an ideal gas, $\Delta T = 0$, so:
$$\Delta U = nC_v\Delta T = \mathbf{0\ J}$$
2. **Reversible Work Done (w):**
$$w = -2.303nRT\log_{10}\left(\frac{P_1}{P_2}\right)$$
$$w = -2.303 \times 2 \times 8.314 \times 300 \times \log_{10}\left(\frac{10}{2}\right)$$
$$w = -11488.3 \times \log_{10}(5) = -11488.3 \times 0.69897 = \mathbf{-8030.0\ J = -8.03\ kJ}$$
(Negative sign indicates work is done BY the system on surroundings).
3. **Heat Absorbed (q):** From First Law: $\Delta U = q + w \Longrightarrow q = \Delta U - w = 0 - (-8030\ \text{J}) = \mathbf{+8030\ J} = \mathbf{+8.03\ kJ}$.

5. The Second Law of Thermodynamics and Entropy

5.1 Limitations of the First Law

The First Law states that energy is conserved in any process, but it fails to answer:
1. **Direction of Spontaneity:** Why heat flows spontaneously from a hot to a cold body, but never spontaneously from cold to hot.
2. **Completeness of Conversion:** Why mechanical work can be completely converted into heat (e.g., via friction), but heat cannot be completely converted into mechanical work without leaving changes in the surroundings.

5.2 Classical Statements of the Second Law

1. **Kelvin-Planck Statement:** "It is impossible for any cyclic device to absorb heat from a single thermal reservoir and convert it completely into equivalent mechanical work without rejecting some waste heat to a lower-temperature sink." (A 100% efficient heat engine is impossible).
2. **Clausius Statement:** "It is impossible to construct a cyclic machine whose sole effect is the transfer of heat from a cooler body to a hotter body without an input of external mechanical work." (Heat cannot spontaneously flow uphill).

5.3 The Concept of Entropy (S)

Entropy is a thermodynamic state function that quantifies the degree of molecular disorder, randomness, or thermal energy dispersal in a system. For a reversible process at absolute temperature T:
$$dS = \frac{dq_{\text{rev}}}{T}$$

- **Units:** J K^{-1} or J $\text{K}^{-1}\text{mol}^{-1}$ (molar entropy).

5.4 Clausius Inequality and Entropy Changes

- For any cyclic process:

$$\oint \frac{dq}{T} \leq 0 \quad (= \text{reversible}, < \text{irreversible})$$

- **For an isolated system ($dq = 0$):**

$$dS_{\text{isolated}} \geq 0$$

- **Total Entropy Change of the Universe (ΔS_{total}):**

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

  - Reversible Process: $\Delta S_{\text{total}} = 0$.
  - Spontaneous (Irreversible) Process: $\Delta S_{\text{total}} > 0$.
  - Impossible Process: $\Delta S_{\text{total}} < 0$.

5.5 Entropy Change in Isothermal Expansion of a Perfect Gas

For an isothermal expansion from V_1 to V_2 at temperature T:

1. **System Entropy Change:** Since $dq_{\text{rev}} = -dw_{\text{rev}} = PdV = \left(nR\frac{T}{V}\right)dV$:

$$\Delta S_{\text{sys}} = \int_{V_1}^{V_2} \frac{dq_{\text{rev}}}{T} = \int_{V_1}^{V_2} \frac{nR}{V} dV = nR \ln\left(\frac{V_2}{V_1}\right)$$

2. **Surroundings Entropy Change (Irreversible expansion against P_{ext}):** $\Delta S_{\text{surr}} = -\frac{q_{\text{actual}}}{T} = -\frac{P_{\text{ext}}(V_2 - V_1)}{T}$.
3. **Free Expansion ($P_{\text{ext}} = 0$):** $q = 0 \Longrightarrow \Delta S_{\text{surr}} = 0$. $\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = nR \ln\left(\frac{V_2}{V_1}\right) + 0 > 0$ (Confirms spontaneity).

6. The Carnot Cycle and Heat Engines

A heat engine is a thermodynamic device that cyclically absorbs thermal energy from a high-temperature source, converts a fraction into mechanical work, and rejects the remainder to a low-temperature sink.

6.1 Architecture and 4-Stroke Cycle

The Carnot cycle consists of 4 reversible operations using 1 mole of an ideal gas inside a cylinder fitted with a frictionless, non-conducting piston:

Carnot Engine PV Indicator Diagram (Schematic Representation):

$$W = Q_H - Q_C$$

$$\eta = \frac{W}{Q_H} = \frac{Q_H - Q_C}{Q_H}$$

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H}$$

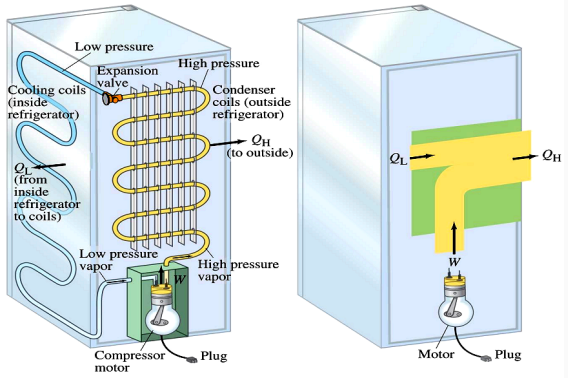


Figure 1.1: (Left) Working principle of cyclic Carnot heat engine; (Right) Official PV Indicator Diagram across Strokes AB, BC, CD, and DA.

Stroke 1 (Curve AB): Reversible Isothermal Expansion at source temperature T_h.

- Heat absorbed from source: $q_h = -w_1 = RT_h \ln\left(\frac{V_2}{V_1}\right)$.

Stroke 2 (Curve BC): Reversible Adiabatic Expansion from T_h to T_c.

- Heat exchange: $q = 0$. Temperature drops to T_c. Work done by gas: $w_2 = C_v(T_c - T_h)$.

Stroke 3 (Curve CD): Reversible Isothermal Compression at sink temperature T_c.

- Heat rejected to sink: $q_c = -w_3 = -RT_c \ln\left(\frac{V_3}{V_4}\right)$. Work done on gas: $w_3 = -RT_c \ln\left(\frac{V_3}{V_4}\right)$.

Stroke 4 (Curve DA): Reversible Adiabatic Compression restoring initial state (P_1, V_1, T_h).

- Heat exchange: $q = 0$. Work done on gas: $w_4 = C_v(T_h - T_c)$.

6.2 Net Cyclic Work and Efficiency Derivation

Total net work done by the engine in one complete cycle:

$$w_{\text{net}} = w_1 + w_2 + w_3 + w_4$$

Notice that $w_2 + w_4 = C_v(T_c - T_h) + C_v(T_h - T_c) = 0$. Therefore:

$$w_{\text{net}} = -RT_h \ln\left(\frac{V_2}{V_1}\right) - RT_c \ln\left(\frac{V_4}{V_3}\right)$$

From adiabatic relations for curves BC and DA:

$$T_h V_2^{\gamma-1} = T_c V_3^{\gamma-1} \quad \text{and} \quad T_h V_1^{\gamma-1} = T_c V_4^{\gamma-1}$$

Dividing the two equations:

$$\left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{V_3}{V_4}\right)^{\gamma-1} \implies \frac{V_2}{V_1} = \frac{V_3}{V_4}$$

Substituting into the work equation:

$$w_{\text{net}} = -R(T_h - T_c) \ln\left(\frac{V_2}{V_1}\right)$$

The thermal efficiency η is defined as the fraction of absorbed heat converted into useful work:

$$\eta = \frac{|w_{\text{net}}|}{q_h} = \frac{R(T_h - T_c)\ln\left(\frac{V_2}{V_1}\right)}{RT_h \ln\left(\frac{V_2}{V_1}\right)} = \frac{T_h - T_c}{T_h} = 1 - \frac{T_c}{T_h}$$

6.3 Key Thermodynamic Deductions from Carnot Efficiency

1. **Temperature Limit:** Since absolute zero ($T_c = 0$ K) is unattainable by Third Law, $\eta < 1$ (100% efficiency is physically impossible).

2. **Working Substance Independence:** The efficiency depends strictly on the operating temperatures of the hot source (T_h) and cold sink (T_c), completely independent of the working substance (Carnot's Theorem).
3. **Coefficient of Performance (COP) for Refrigerators and Heat Pumps:**
   - **Refrigerator COP (β):** Measures cooling efficiency:

$$\text{COP}_{\text{ref}} = \frac{q_c}{|w_{\text{net}}|} = \frac{T_c}{T_h - T_c}$$

   - **Heat Pump COP:** Measures heating efficiency:

$$\text{COP}_{\text{HP}} = \frac{q_h}{|w_{\text{net}}|} = \frac{T_h}{T_h - T_c} = \text{COP}_{\text{ref}} + 1$$

[FAT Comprehensive 10-Mark Exam Question]
Question: With the aid of a neat PV indicator diagram, describe the four stages of a Carnot cycle and derive the expression for its thermal efficiency. A Carnot heat engine operates between 800 K and 300 K. If it absorbs 2400 J of heat from the source per cycle, calculate: (i) Thermal efficiency, (ii) Work done per cycle, and (iii) Heat rejected to the sink.
Step-by-Step Scoring Solution:
1. **Diagram & 4 Stages Description:** Draw PV curves AB, BC, CD, DA. Explain strokes 1 to 4 with heat and work expressions. (4 Marks)
2. **Efficiency Derivation:** Derive $w_{\text{net}} = -R(T_h - T_c)\ln\left(\frac{V_2}{V_1}\right)$ and $\eta = \frac{T_h - T_c}{T_h}$. (3 Marks)
3. **Numerical Calculation:** (3 Marks)
   - $\eta = 1 - \frac{T_c}{T_h} = 1 - \frac{300}{800} = 1 - 0.375 = \mathbf{0.625}$ $\quad(\mathbf{62.5\%})$.
   - $|w_{\text{net}}| = \eta \times q_h = 0.625 \times 2400\ \text{J} = \mathbf{1500\ J}$.
   - $q_c = q_h - |w_{\text{net}}| = 2400 - 1500 = \mathbf{900\ J}$.

7. Free Energy and Chemical Spontaneity

7.1 Helmholtz Free Energy (A) and Gibbs Free Energy (G)

- **Helmholtz Free Energy ($A = U - TS$):** Measures maximum total work obtainable from a system in a reversible isothermal process ($w_{\text{max}} = -\Delta A$).
- **Gibbs Free Energy ($G = H - TS$):** Measures maximum non-expansion (useful electrical/chemical) work obtainable at constant temperature and pressure:

$$w_{\text{useful}} = -\Delta G$$

7.2 Derivation of the Gibbs-Helmholtz Equation

By definition: $G = H - TS$. Taking the total differential:

$$dG = dH - TdS - SdT$$

Since $H = U + PV$, $dH = dU + PdV + VdP$. From First Law, $dU = dq - PdV = TdS - PdV$ (for a reversible process). Substituting into dH: $dH = TdS + VdP$. Therefore:

$$dG = (TdS + VdP) - TdS - SdT = VdP - SdT$$

At constant pressure ($dP = 0$):

$$\left(\frac{\partial G}{\partial T}\right)_P = -S$$

For a finite process: $\left(\partial \frac{\Delta G}{\partial} T\right)_P = -\Delta S$. Substituting $-\Delta S = \frac{\Delta G - \Delta H}{T}$ yields the standard **Gibbs-Helmholtz Equation:**

$$\Delta G = \Delta H + T\left(\frac{\partial(\Delta G)}{\partial T}\right)_P$$

Or in its equivalent derivative quotient form:

$$\left[\frac{\partial}{\partial T}\left(\frac{\Delta G}{T}\right)\right]_P = -\frac{\Delta H}{T^2}$$

7.3 Spontaneity Criteria and the 4 Thermodynamic Regimes

Under constant temperature and pressure:

$$\Delta G = \Delta H - T\Delta S$$

- **Spontaneous Reaction:** $\Delta G < 0$ (Exergonic process; system releases free energy).
- **Equilibrium State:** $\Delta G = 0$ (No net driving force).
- **Non-Spontaneous Reaction:** $\Delta G > 0$ (Endergonic process; forward reaction requires continuous energy input).

ΔH	ΔS	**Spontaneity Regime (ΔG)**	**Example**
$-$ (Exo)	$+$ (Disorder)	**Always Spontaneous ($\boldsymbol{\Delta G < 0}$)** at all temperatures.	Combustion of propane; decomposition of H_2O_2
$+$ (Endo)	$-$ (Order)	**Always Non-Spontaneous ($\boldsymbol{\Delta G > 0}$)** at all temperatures.	Photosynthesis ($6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$)
$-$ (Exo)	$-$ (Order)	**Spontaneous at LOW temperatures** ($\lvert\Delta H\rvert > \lvert T\Delta S\rvert$); non-spontaneous at high T.	Haber synthesis ($N_2 + 3H_2 \rightarrow 2NH_3$); freezing of water
$+$ (Endo)	$+$ (Disorder)	**Spontaneous at HIGH temperatures** ($\lvert T\Delta S\rvert > \lvert\Delta H\rvert$); non-spontaneous at low T.	Thermal decomposition of $CaCO_3$; vaporization of water

Crossover Equilibrium Temperature ($\boldsymbol{T_{eq}}$):
When $\Delta G = 0$:

$$0 = \Delta H - T_{eq}\Delta S \Longrightarrow \boldsymbol{T_{eq} = \frac{\Delta H}{\Delta S}}$$

8. The Third Law of Thermodynamics and Residual Entropy

Third Law (Nernst Heat Theorem / Planck Statement):
"The entropy of every pure, perfectly crystalline substance approaches zero as the absolute temperature approaches zero Kelvin (0 K):"

$$\lim_{T \to 0} S = 0$$

8.1 Determination of Absolute Entropies

Because the absolute entropy is zero at 0 K, the absolute molar entropy S_T of a substance at any temperature T can be directly calculated by integrating its heat capacity:

$$S_T = \int_0^T \frac{C_p}{T} dT = \int_0^T C_p d(\ln T)$$

If phase transitions occur between 0 K and T, latent heat terms ($\Delta \frac{H_{trans}}{T_{trans}}$) are added.

8.2 Residual Entropy

A pure substance that does not form a perfect crystal at 0 K retains a non-zero entropy called **residual entropy**, caused by structural or orientational disorder frozen into the crystal lattice:

$$S_{\text{residual}} = k_B \ln \Omega$$

where k_B is Boltzmann's constant and Ω is the number of accessible microstates.

- **Carbon Monoxide (CO):** In the solid lattice, dipole moment is very small (0.11 D). Dipoles freeze in random $C-O$ and $O-C$ orientations ($\Omega = 2^N$), giving:

$$S_{\text{residual}} = R \ln 2 \approx \mathbf{5.76 J\ K^{-1} mol^{-1}}$$

- **Nitrous Oxide (N_2O):** Linear molecule freezes randomly as $N-N-O$ or $O-N-N$ ($S_{\text{residual}} \approx 5.8 \text{J K}^{-1}\text{mol}^{-1}$).
- **Water Ice (H_2O):** Residual entropy arises from hydrogen bonding arrangement disorder (Pauling calculated $S_{\text{residual}} = R \ln\left(\frac{3}{2}\right) \approx \mathbf{3.37 J\ K^{-1} mol^{-1}}$).

PART B: CHEMICAL KINETICS

9. Reaction Rates, Rate Laws, Order, and Molecularity

Chemical kinetics investigates the speeds or rates of chemical reactions, the factors influencing these rates, and the molecular mechanisms involved.

9.1 Rate of Reaction

For a general reaction: $aA + bB \rightarrow cC + dD$ The instantaneous rate of reaction is defined as:

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt}$$

- **Negative sign:** Reactant concentrations decrease over time.
- **Positive sign:** Product concentrations increase over time.

9.2 The Differential Rate Law

$$\text{Rate} = k[A]^m[B]^n$$

where:

- k is the specific reaction rate constant.
- m is the partial order with respect to reactant A.
- n is the partial order with respect to reactant B.
- $n_{\text{total}} = m + n$ is the overall order of the reaction.

Critical Note: Reaction orders (m, n) are experimentally determined. They CANNOT be deduced from stoichiometric coefficients (a, b) of the balanced equation.

9.3 Comprehensive Comparison: Order vs. Molecularity

Feature	Order of Reaction	Molecularity of Reaction
Definition	Sum of powers of concentration terms in the experimental rate law.	Number of reacting species (atoms/ions/molecules) colliding simultaneously in an elementary step.
Determination	Purely experimental quantity; cannot be guessed from stoichiometry.	Theoretical quantity deduced from the mechanism of an elementary step.
Values Possible	Can be zero, positive, negative, or fractional $(0, 1, 2, 3, \frac{1}{2}, \frac{3}{2})$.	Can ONLY be positive whole integers $(1, 2, 3)$. Cannot be zero or fractional.
Applicability	Applies to both elementary and complex multi-step reactions.	Applies ONLY to individual elementary steps; meaningless for overall complex reactions.

Pressure/Temp Dependence	Can change with experimental temperature, pressure, or concentration.	Constant and characteristic for a given elementary reaction step.

10. Integrated Rate Equations and Half-Life Derivations

Order	Differential Law	Integrated Rate Equation	Half-Life ($t_{\frac{1}{2}}$)	Units of k
0	$-d\frac{[A]}{d}t = k$	$[A]_t = [A]_0 - kt$	$\frac{[A]_0}{2k}$	mol $L^{-1}s^{-1}$
1	$-d\frac{[A]}{d}t = k[A]$	$\ln\left(\frac{[A]_0}{[A]_t}\right) = kt \Longrightarrow k = \frac{2.303}{t}\log_{10}\left(\frac{[A]_0}{[A]_t}\right)$	$\frac{\ln 2}{k} = \frac{0.693}{k}$	s^{-1}
2	$-d\frac{[A]}{d}t = k[A]^2$	$\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$	$\frac{1}{k[A]_0}$	L $mol^{-1}s^{-1}$

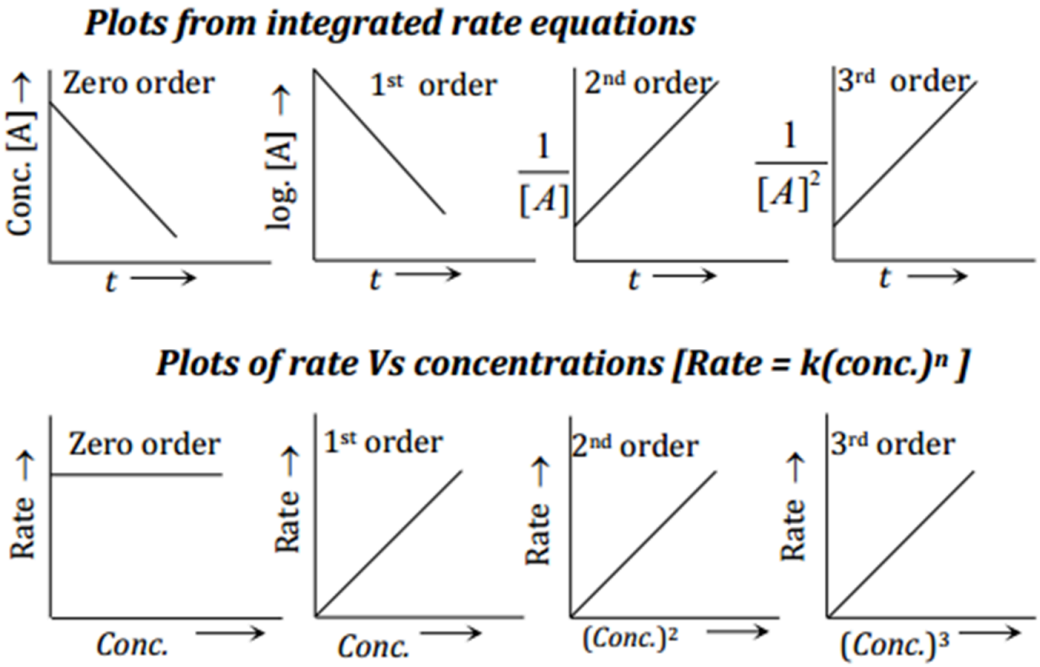


Figure 1.4: Linear graphical plots derived from integrated rate equations: (Left) Zero-order $[A]_t$ vs t; (Center) First-order $\ln[A]_t$ vs t; (Right) Second-order $\frac{1}{[A]_t}$ vs t.

10.1 Step-by-Step Derivation: First-Order Reaction

1. Differential rate expression:

$$-\frac{d[A]}{dt} = k[A] \Longrightarrow \frac{d[A]}{[A]} = -kdt$$

2. Integrating both sides between limits $t = 0$ ($[A] = [A]_0$) and $t = t$ ($[A] = [A]_t$):

$$\int_{[A]_0}^{[A]_t} \frac{d[A]}{[A]} = -k\int_0^t dt$$

$$[\ln[A]]_{[A]_0}^{[A]_t} = -kt \Longrightarrow \ln[A]_t - \ln[A]_0 = -kt$$

3. Rearranging into standard logarithmic forms:

$$\ln[A]_t = -kt + \ln[A]_0 \quad (\text{Plot of } \ln[A]_t \text{ vs } t \text{ is linear with slope } -k)$$

$$k = \frac{1}{t}\ln\left(\frac{[A]_0}{[A]_t}\right) = \mathbf{\frac{2.303}{t}\log_{10}\left(\frac{[A]_0}{[A]_t}\right)}$$

4. **Half-Life ($t_{\frac{1}{2}}$):** Time when $[A]_t = \frac{[A]_0}{2}$:

$$t_{\frac{1}{2}} = \frac{2.303}{k}\log_{10}\left(\frac{[A]_0}{\frac{[A]_0}{2}}\right) = \frac{2.303}{k}\log_{10}(2) = \frac{2.303 \times 0.30103}{k} = \mathbf{\frac{0.693}{k}}$$

Crucial Exam Fact: The half-life of a first-order reaction is completely independent of the initial reactant concentration $[A]_0$.

10.2 Pseudo-First-Order Reactions

A reaction that is bimolecular or second-order overall, but conforms experimentally to first-order kinetics because one of the reactants is present in large stoichiometric excess, keeping its concentration virtually constant.

- **Example 1: Acid Hydrolysis of Methyl Acetate:**

$$CH_3COOCH_3 + H_2O(\text{excess}) \xrightarrow{H^+} CH_3COOH + CH_3OH$$

  Rate $= k'[CH_3COOCH_3][H_2O]$. Since $[H_2O] \gg [CH_3COOCH_3]$, $[H_2O] \approx$ constant. Rate $= k[CH_3COOCH_3]$, where $k = k'[H_2O]$ (pseudo-first-order rate constant).
- **Example 2: Inversion of Cane Sugar:**

$$C_{12}H_{22}O_{11} + H_2O(\text{excess}) \xrightarrow{H^+} C_6H_{12}O_6(\text{glucose}) + C_6H_{12}O_6(\text{fructose})$$

[FAT Exam Numerical Problem — 5 Marks]

Problem: A first-order reaction is 50% complete in 25 minutes at 300 K. Calculate: (i) the specific rate constant k, and (ii) the time required for 80% completion of the reaction.

Step-by-Step Solution:

1. **Calculate k from half-life:**

$$k = \frac{0.693}{t_{\frac{1}{2}}} = \frac{0.693}{25\text{ min}} = \mathbf{0.02772 min^{-1}} = \mathbf{4.62 \times 10^{-4} s^{-1}}$$

2. **Calculate time for 80% completion:** When reaction is 80% complete, remaining reactant is $[A]_t = [A]_0 - 0.80[A]_0 = 0.20[A]_0$.

$$t = \frac{2.303}{k}\log_{10}\left(\frac{[A]_0}{0.20[A]_0}\right) = \frac{2.303}{0.02772}\log_{10}(5)$$

$$t = \frac{2.303}{0.02772} \times 0.69897 = 83.08 \times 0.69897 = \mathbf{58.07\text{ minutes}}$$

11. Collision Theory and Reaction Dynamics

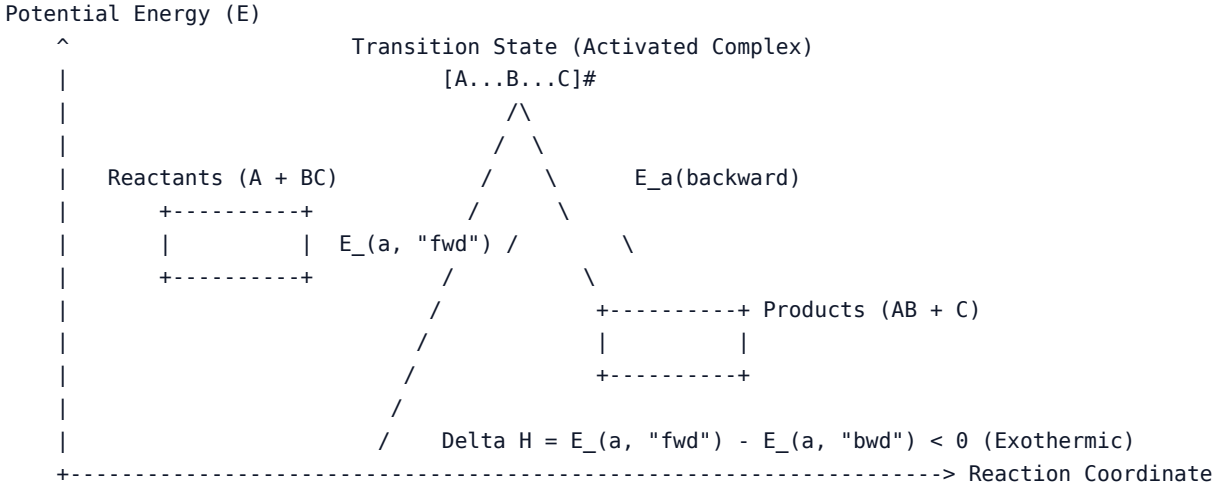
11.1 Postulates of Collision Theory

Developed by Max Trautz and William Lewis:

1. Chemical reactions occur as a result of physical collisions between reacting molecules.
2. The rate of reaction is directly proportional to the collision frequency Z (number of collisions per unit volume per unit time).
3. Not all collisions lead to chemical reaction. Only **effective collisions** produce products. An effective collision requires:
   - **Energy Factor (Activation Energy Barrier E_a):** Colliding molecules must possess a minimum threshold energy E_{th}. Fraction of molecules with $E \geq E_a$ is given by Boltzmann factor: $f = e^{-\frac{E_a}{R}T}$.
   - **Orientation Factor (Steric Factor P):** Reacting molecules must collide with the proper spatial geometric orientation so that appropriate bonds break and form.
4. **Rate Equation from Collision Theory:**

$$k = P \times Z \times e^{-\frac{E_a}{RT}}$$

11.2 Reaction Coordinate Energy Profiles



- **Activation Energy (E_a):** The minimum excess energy above the ground state that reactant molecules must acquire to form the activated transition state:

$$E_a = E_{\text{threshold}} - E_{\text{reactants}}$$

- **For an Exothermic Reaction:** $E_{\text{products}} < E_{\text{reactants}} \implies \Delta H = E_{a,\text{ fwd}} - E_{a,\text{ bwd}} < 0$.
- **For an Endothermic Reaction:** $E_{\text{products}} > E_{\text{reactants}} \implies \Delta H = E_{a,\text{ fwd}} - E_{a,\text{ bwd}} > 0$.

12. Effect of Temperature and the Arrhenius Equation

In 1889, Svante Arrhenius formulated the quantitative mathematical relation governing the temperature dependence of reaction rates:

$$k = Ae^{-\frac{E_a}{RT}}$$

where:

- k is the reaction rate constant.
- A is the Arrhenius pre-exponential factor (frequency factor, same units as k).
- E_a is the activation energy (J mol^{-1} or kJ mol^{-1}).
- $R = 8.314 \text{J K}^{-1}\text{mol}^{-1}$ is the universal gas constant.
- T is the absolute temperature in Kelvin.

12.1 Logarithmic Forms and Graphical Determination

Taking natural logarithms of both sides:

$$\ln k = \ln A - \frac{E_a}{RT}$$

Converting to common logarithm (base 10):

$$\log_{10} k = \log_{10} A - \frac{E_a}{2.303RT}$$

- **Arrhenius Plot:** A plot of $\log_{10} k$ against $\frac{1}{T}$ produces a straight line:
  - Slope $= -\frac{E_a}{2.303R} \implies \boldsymbol{E_a = -2.303 \times R \times \text{Slope}}$
  - Y-intercept $= \log_{10} A$

12.2 Two-Temperature Formula

Evaluating the Arrhenius equation at two different temperatures T_1 and T_2 ($T_2 > T_1$) with rate constants k_1 and k_2:

$$\ln k_1 = \ln A - \frac{E_a}{RT_1} \quad \text{and} \quad \ln k_2 = \ln A - \frac{E_a}{RT_2}$$

Subtracting the first equation from the second:

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] = \frac{E_a}{R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

In common logarithm:

$$\boldsymbol{\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]}$$

[High-Frequency FAT Exam Numerical Problem — 10 Marks]

Problem: The rate constant of a reaction is $1.60 \times 10^{-3}\text{s}^{-1}$ at 300 K and increases to $6.40 \times 10^{-3}\text{s}^{-1}$ at 320 K. Calculate: (i) the activation energy (E_a) of the reaction, and (ii) the rate constant at 350 K. ($R = 8.314\text{J K}^{-1}\text{mol}^{-1}$).

Step-by-Step Model Scoring Solution:

1. **Calculate Activation Energy (E_a):**

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303 \times 8.314}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

$$\log_{10}\left(\frac{6.40 \times 10^{-3}}{1.60 \times 10^{-3}}\right) = \frac{E_a}{19.147}\left[\frac{320 - 300}{300 \times 320}\right]$$

$$\log_{10}(4) = \frac{E_a}{19.147}\left[\frac{20}{96000}\right] \Longrightarrow 0.60206 = \frac{E_a}{19.147} \times 2.0833 \times 10^{-4}$$

$$E_a = \frac{0.60206 \times 19.147}{2.0833 \times 10^{-4}} = \frac{11.5276}{2.0833 \times 10^{-4}} = \mathbf{55333\,J\ mol^{-1} = 55.33\,kJ\ mol^{-1}}$$

2. **Calculate Rate Constant at $T_3 = 350$ K (k_3):**

$$\log_{10}\left(\frac{k_3}{k_1}\right) = \frac{55333}{19.147}\left[\frac{350 - 300}{300 \times 350}\right] = 2890.0 \times \frac{50}{105000} = 2890.0 \times 4.7619 \times 10^{-4} = 1.3762$$

$$\frac{k_3}{k_1} = 10^{1.3762} = 23.78$$

$$k_3 = 23.78 \times k_1 = 23.78 \times (1.60 \times 10^{-3}\text{s}^{-1}) = \mathbf{3.80 \times 10^{-2}\,s^{-1}}$$

13. Catalysis and Mechanism of Rate Enhancement

A catalyst is a chemical substance that increases the rate of a chemical reaction without undergoing any permanent chemical or stoichiometric change itself.

13.1 Mechanism of Catalytic Action

- A catalyst provides an **alternative reaction pathway** with a significantly lower activation energy ($E_{a'} < E_a$).
- Because the exponential factor in the Arrhenius equation is $e^{-\frac{E_a}{R}T}$, a modest decrease in E_a leads to an exponential increase in the fraction of molecules possessing sufficient kinetic energy to react.
- **Crucial Thermodynamic Invariants:**
  1. A catalyst **does NOT change** the initial reactant energy or final product energy.
  2. Enthalpy change (ΔH), Gibbs free energy (ΔG), and equilibrium constant (K_{eq}) remain strictly unchanged.
  3. A catalyst speeds up both forward and reverse reaction rates by the exact same factor, thereby helping the system achieve dynamic equilibrium faster without altering equilibrium yield.

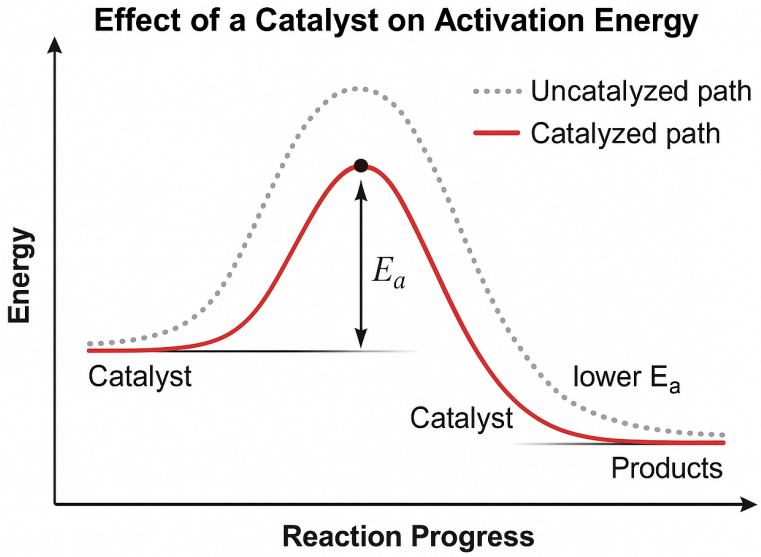


Figure 1.3: Potential energy diagram comparing the uncatalyzed reaction path (E_a) with the catalyzed path ($E_{a'}$) showing lowered activation energy barrier.

13.2 Homogeneous vs. Heterogeneous Catalysis

Parameter	Homogeneous Catalysis	Heterogeneous Catalysis
Phase Relationship	Catalyst and reactants exist in the same physical phase (all liquid or all gas).	Catalyst and reactants exist in different physical phases (typically solid catalyst with gaseous or liquid reactants).
Mechanism	Formation of an intermediate coordination complex throughout the bulk solution.	Surface adsorption (physisorption followed by chemisorption), reaction on active sites, and desorption.
Industrial Separation	Difficult and costly to separate catalyst from reaction products.	Simple separation via filtration, centrifugation, or fixed-bed reactors.
Industrial Example 1	Gas-phase decomposition of ozone catalyzed by chlorine radicals: $O_3 + Cl^{\cdot} \rightarrow ClO^{\cdot} + O_2$.	Haber Process for ammonia synthesis: solid porous Iron promoted with K_2O and Al_2O_3: $N_2(g) + 3H_2(g) \overset{\text{Fe}}{\rightarrow} 2NH_3(g)$.
Industrial Example 2	Acid-catalyzed ester hydrolysis in aqueous solution: $CH_3COOR + H_2O \overset{H^+}{\rightarrow} CH_3COOH + ROH$.	Contact Process for sulfuric acid: oxidation of SO_2 over solid Vanadium pentoxide: $2SO_2(g) + O_2(g) \overset{V_2O_5}{\rightarrow} 2SO_3(g)$.
Industrial Example 3	Lead Chamber Process: oxidation of SO_2 catalyzed by gaseous nitric oxide: $2SO_2 + O_2 \overset{\text{NO}}{\rightarrow} 2SO_3$.	Ostwald Process for nitric acid: oxidation of ammonia over Platinum-Rhodium gauze: $4NH_3(g) + 5O_2(g) \overset{\text{Pt-Rh}}{\rightarrow} 4NO(g) + 6H_2O(g)$.